

UNIVERSITY OF CHITTAGONG
CSE 714 · AI Lab

CROPDX



CropDiseaseDiagnosisExpertSystem — Implemented in Turbo Prolog

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Expert System

Turbo Prolog

Rule-Based AI

Agricultural AI

Backward Chaining

4
CROPS COVERED

8
DISEASES

24
DIAGNOSTIC RULES

6
QUESTIONS/CROP

∞
OFFLINE — NO INTERNET

ABSTRACT

CROPDX is a rule-based expert system implemented in Turbo Prolog that simulates the diagnostic reasoning of an experienced agronomist. It supports four staple crops — Rice, Potato, Tomato, and Wheat — and identifies eight distinct diseases through a structured yes/no symptom questionnaire, returning confirmed diagnosis, causative organism, treatment protocol, and preventive practices.

PROBLEM STATEMENT

Crop diseases are a leading cause of yield loss in developing economies. In rural Bangladesh, farmers lack timely access to plant pathologists — delayed diagnosis allows diseases to spread and devastate harvests.

Geographical Isolation

Experts concentrated in urban centres; rural access minimal.

Symptom Misidentification

Many diseases share visually similar symptoms (e.g. Rice Blast vs. Bacterial Leaf Blight).

No Digital Tools

Internet connectivity in rural areas remains unreliable — cloud solutions impractical.

Pesticide Overuse

Without accurate diagnosis, broad-spectrum pesticides applied unnecessarily.

SYSTEM ARCHITECTURE

User Interface (CLI)
write/readln predicates — yes/no symptom elicitation

Inference Engine
Prolog Backward-Chaining via top-down clause resolution

Knowledge Base
24 diagnostic rules encoded in Prolog clause format

Output Module — show_result/4
Disease · Cause · Treatment · Prevention

METHODOLOGY

Knowledge Engineering methodology in 5 phases:

1 Domain Analysis

Identification of 8 diagnostically distinct diseases.

2 Knowledge Acquisition

Extraction of symptom sets, pathogens, and treatment protocols.

3 Knowledge Representation

Encoding rules in Turbo Prolog clause format.

4 Implementation

CROPDX.PRO compiled and tested in DOSBox/Turbo Prolog.

5 Validation

Inference paths traced for all 8 diseases to verify correctness.

CROP-DISEASE KNOWLEDGE MAP

RICE

Rice Blast

Bacterial Leaf Blight

POTATO

Late Blight

Early Blight

TOMATO

Mosaic Virus

Fusarium Wilt

WHEAT

Yellow Rust

Powdery Mildew

CROPDX Expert System

ECONOMIC IMPACT OF DISEASES

Disease	Impact
Rice Blast	Up to 30% yield loss annually worldwide
Late Blight	Caused Irish Famine (1845); still devastating
Early Blight	20–30% yield reduction in unmanaged fields
Yellow Rust	Global wheat losses \$1B+ annually

KEY REFERENCES

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[5] Ou, S. H. (1985). Rice Diseases, 2nd ed. CAB International.

[6] Agrios, G. N. (2005). Plant Pathology, 5th ed.Academic Press.

[7] Strange, R. N. and Scott, P. R. (2005). Plant disease: A threat to global food security. Annual Review of Phytopathology, 43, 83–116.

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Scan for Source Code & Project Files



KNOWLEDGE BASE — DISEASE-SYMPTOM MAPPING

Crop	Disease	Pathogen	Key Symptoms
RICE	Rice Blast	Magnaporthe oryzae	Diamond spots, gray centre, brown border
RICE	Bacterial Leaf Blight	Xanthomonas oryzae	Yellow wilting tips, water-soaked streaks
POTATO	Late Blight	Phytophthora infestans	Dark water-soaked patches, white underleaf mold
POTATO	Early Blight	Alternaria solani	Brown spots, yellow halo, lower leaves first
TOMATO	Mosaic Virus	Tomato Mosaic Virus	Mosaic pattern, leaf curl, stunted growth
TOMATO	Fusarium Wilt	Fusarium oxysporum	Lower wilting, night recovery, brown stem
WHEAT	Yellow Rust	Puccinia striiformis	Yellow pustules in rows, pale yellowing
WHEAT	Powdery Mildew	Blumeria graminis	White powdery patches, easily rubbed off

DIAGNOSTIC RULE PATTERN — 6-SYMPTOM STRUCTURE

Disease 1

yes yes yes _ _ _

S1 S2 S3 S4 S5 S6

Disease 2

_ _ _ yes yes yes

Required symptom Wildcard (_ = any answer)

Each disease requires exactly 3 consecutive "yes" answers. The underscore wildcard ignores the remaining 3 symptoms — enabling clean, robust pattern-matching with no overlap between diseases.

PROLOG IMPLEMENTATION — CORE DIAGNOSTIC RULE

```
/* Rule 1: Rice Blast — first 3 symptoms
YES */
diagnose_rice "yes" "yes" "yes" _ _ _
show_result
"RICE BLAST"
"Magnaporthe oryzae"
"Apply Tricyclazole or Carbendazim"
"Use resistant varieties"
!

/* Rule 2: Bacterial Leaf Blight — last 3
YES */
diagnose_rice _ _ _ "yes" "yes" "yes"
show_result "BACTERIAL LEAF BLIGHT" _ _ _ !

/* Fallback wildcard */
diagnose_rice _ _ _ _ _ _
show_unknown
```

Cut Operator (!): Placed after each successful show_result/4 call. Without it, Prolog backtracks after printing the result and incorrectly executes the wildcard clause — producing duplicate output.

SYSTEM FLOWCHART

START

Display Welcome Header and Crop Menu

Valid crop selected? → Show error, retry

Ask 6 Symptom Questions (yes/no)

Pattern matched in knowledge base? → show_unknown

show_result/4 — Disease · Cause · Treatment · Prevention

Diagnose again? (yes/no) → Restart menu

END

ADVANTAGES

Offline Operation

No internet required — any DOS-capable hardware

Transparent Reasoning

Readable Prolog rules, auditable by domain experts

Easily Extensible

New diseases added by appending clauses only

Zero Bias

Uniform, consistent recommendations every time

Low Cost Deployment

DOSBox (free) + standard PC, no subscription needed

LIMITATIONS

Binary Symptoms Only

No severity levels or numeric measurements

Limited Coverage

Only 2 diseases per crop

No Certainty Factors

Partial symptom matches cannot trigger diagnosis

Static Knowledge Base

Cannot learn from new cases; manual update only

FUTURE IMPROVEMENTS

1 Expanded Knowledge Base

20+ diseases per crop using phytopathology databases

2 Certainty Factors

Mycin-style CF values for partial symptom matching

3 Image-Based Diagnosis

CNN leaf classifier (PlantVillage) replacing manual input

4 Bengali Language Support

Symptom questions in native regional languages

Mobile App

Android with offline SQLite knowledge base for field use

BACKWARD-CHAINING INFERENCE — HOW IT WORKS

Goal: diagnose_rice(?)

Try Clause 1: Rice Blast ("yes","yes","yes",_,_,_) Try Clause 2: Leaf Blight (_,_,_,_,_,_)

Unifies? → show_result/4 X Fails → Backtrack

Cut (!) — Commit, stop search Wildcard Fallback show_unknown/Q

Prolog resolves clauses top-down; backtracking explores alternatives automa

TOOLS AND ENVIRONMENT

Component	Details
Language	Turbo Prolog 2.0
Platform	DOSBox 0.74 on Windows 10
IDE	Turbo Prolog Integrated Editor
Compile	Alt+C (internal compiler)
Run	F9 (compile and run)
Source File	CROPDX.PRO

CONCLUSION

CROPDX achieves all stated objectives — functional, extensible, and well-documented. It correctly diagnoses 8 diseases across 4 crops and demonstrates that classical AI paradigms retain significant practical value for offline, resource-constrained agricultural advisory applications.

EXPERT SYSTEM

PROLOG

RULE-BASED AI

BACKWARD CHAINING

KNOWLEDGE BASE

AGRICULTURAL AI

SPECIFIC OBJECTIVES ACHIEVED

#	Objective	Status
1	Encode 4 crops, 8 diseases into Prolog knowledge base	✓ Done
2	User-friendly interactive CLI via write/readln	✓ Done
3	Differentiate diseases with similar symptom profiles	✓ Done
4	Graceful fallback when no pattern is matched	✓ Done
5	Session management — multiple diagnoses per run	✓ Done
6	Turbo Prolog syntax and DOSBox compatibility	✓ Done
7	Structured output: Disease → Cause → Treatment → Prevention	✓ Done

Submitted to: Dr. Mohammad Shahadat Hossain, Professor · Dept. of Computer Science and Engineering · University of Chittagong

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